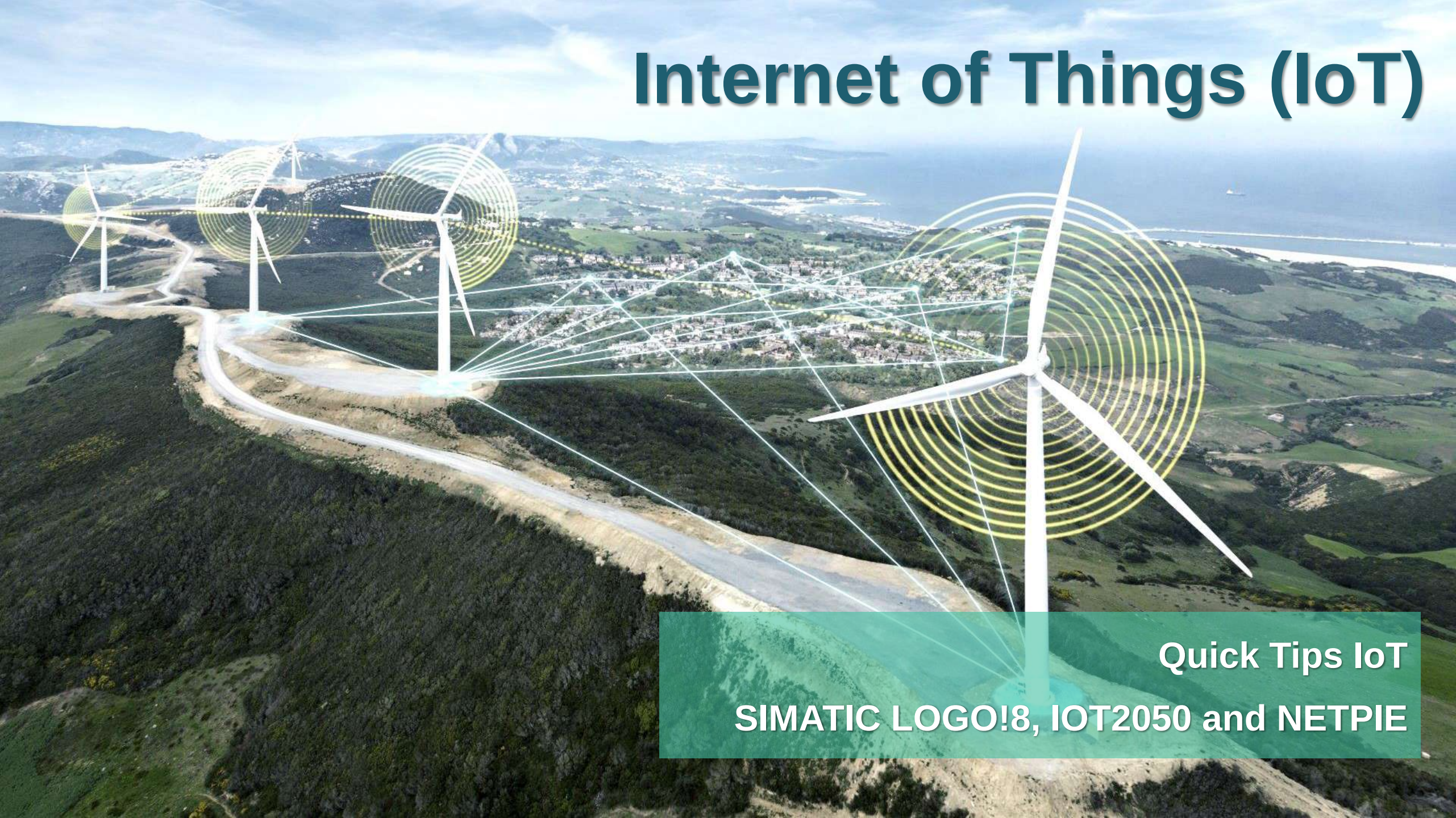


Internet of Things (IoT)



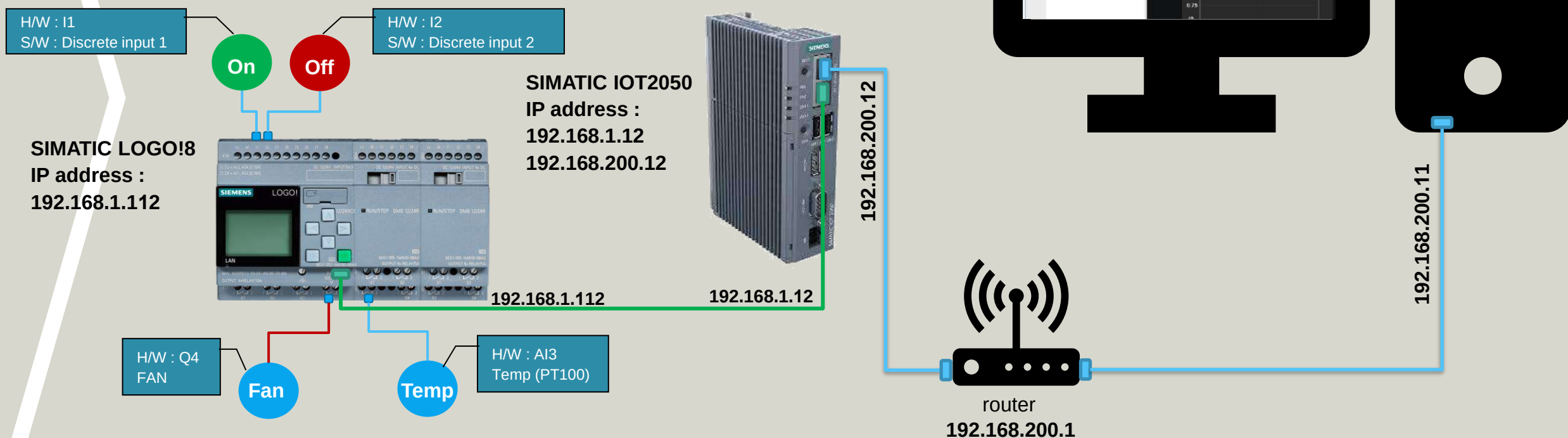
Quick Tips IoT
SIMATIC LOGO!8, IOT2050 and NETPIE

SIMATIC LOGO!8, IOT2050, NETPIE : NODE-Red & MQTT (JSON)



Computer

- LOGO! Soft Comfort V8.3
- NODE-Red (IOT2050)
- NETPIE (Browser)



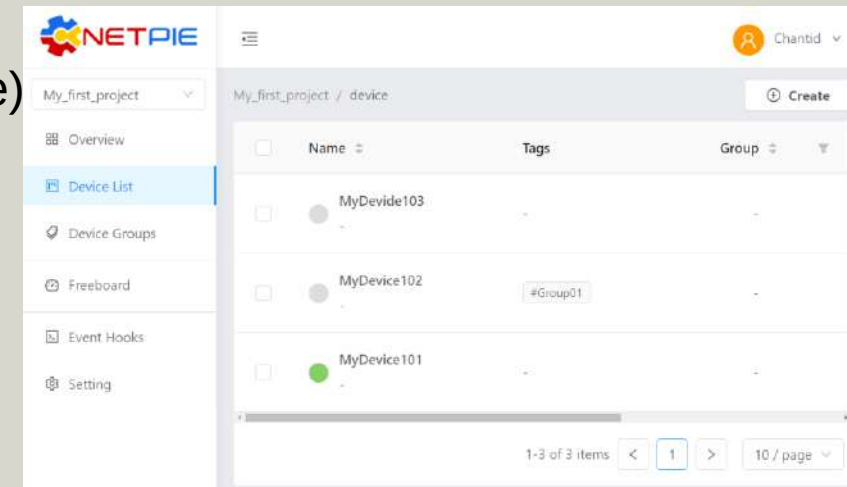
NETPIE2020



Introducing NETPIE 2020

NETPIE 2020 is the latest version of NETPIE. It aims to fulfill the needs of commercial users, especially those in the industrial sector. The platform's new version can shorten and ease the process of IoT product development considerably, from the stage of designing, prototyping, implementing right down to administering and maintaining. NETPIE 2020 comes with the newly designed architecture and many exciting new features.

- 1) Create new device
 - setting the device shadow
- 2) NETPIE freeboard
 - create new Dashboard
 - create new data source
 - add new PAN
 - add new WIDGET (Gauge)



Create new Device : name “MyDevice101”



NETPIE

My_first_project

Overview

Device List

Device Groups

My_first_project / device

Chantid

Create

Name	Tags	Group	Create Date
MyDevide103			
MyDevice102	#Group01		
MyDevice101			

Create

* Name MyDevice101

Description:

Tag: + New Tag

Cancel Create

My_first_project / device

Name	Tags	Group
MyDevide103		
MyDevice102	#Group01	
MyDevice101		

Create Device shadow of “MyDevice101”



NETPIE

My_first_project

Overview

Device List

Device Groups

Freeboard

Event Hooks

Setting

My_first_project / device / MyDevice101

Description

Key

Client ID : [redacted]

Token : [redacted]

Secret : [redacted]

Status : Online

Enable : [toggle]

Shadow Schema Trigger Feed

Code

```
1 {  
2   "data101": 37,  
3   "": null  
4 }
```

ตัวอย่าง JSON object
สำหรับ 1 ข้อมูล ชื่อ
“data101”

```
{ "data101": 37, "": null }
```

Create Dashboard in NETPIE Freeboard



The screenshot illustrates the process of creating a dashboard in the NETPIE Freeboard interface. The interface is divided into a sidebar on the left and a main content area. The sidebar contains the NETPIE logo, a project selector (My_first_project), and a list of navigation items: Overview, Device List, Device Groups, Freeboard (selected), and Event Hooks. The main content area shows a table with columns for Name and Create Date. A 'Create' button is visible in the top right corner of the main area.

A 'Create Dashboard' dialog box is open, showing the following fields:

- * Name: DashBoard101
- Description: (empty text area)
- Buttons: Cancel and Create

Below the dialog box, the main content area shows the updated table with the newly created dashboard:

Name	Create Date	Actions
DashBoard101	2021-08-03 22:14	Edit Delete

The 'Edit' button is highlighted with a red dashed box and a hand cursor, indicating the next step in the process.

Create DATA SOURCES for your Dashboard



 **NETPIE Freeboard**

IMPORT

EXPORT

RESET

+ ADD PANE

SAVE

DATA SOURCES

Name

Data_1

TempData

ADD

DATA SOURCE

NAME

Data_1

DEVICE ID

Client ID ของ Device ที่ต้องการอ่านข้อมูล

DEVICE TOKEN

Token ของ Device ที่ต้องการอ่านข้อมูล

SUBSCRIBED TOPICS

Topic ที่ต้องการ Subscribe

FEED

YES

SINCE

6

Hour

Display data points since ... ago

DOWN SAMPLING

1

Second

Resolution of the data points.

SAVE

CANCEL

Device information

Key

Client ID :

Token :

Secret :

Status :

Online

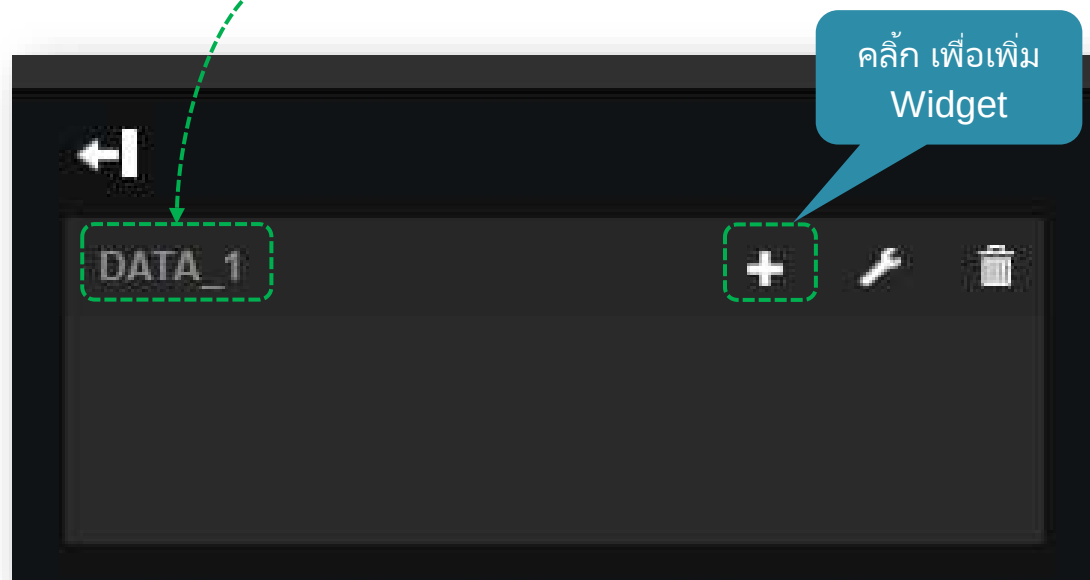
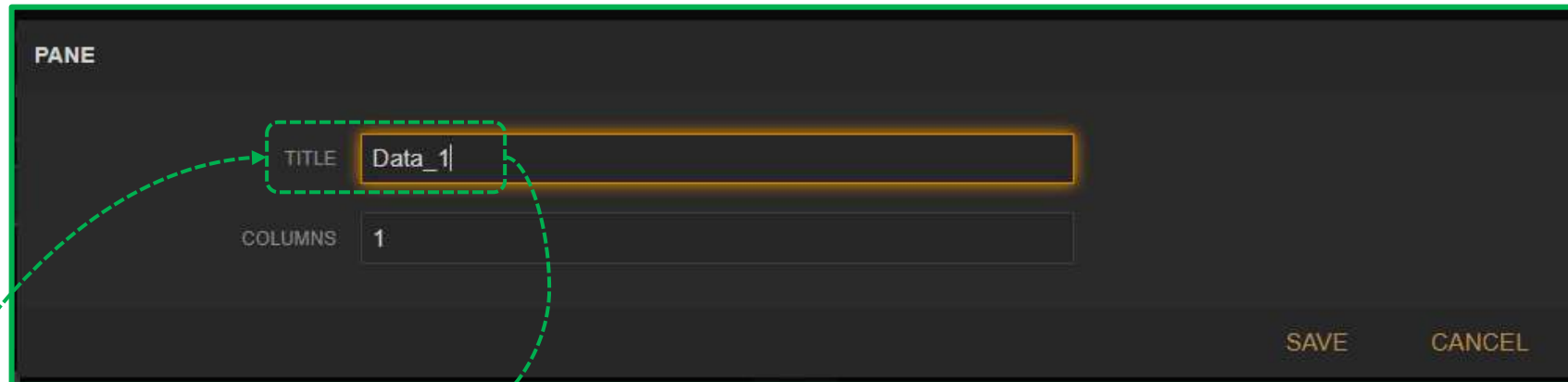
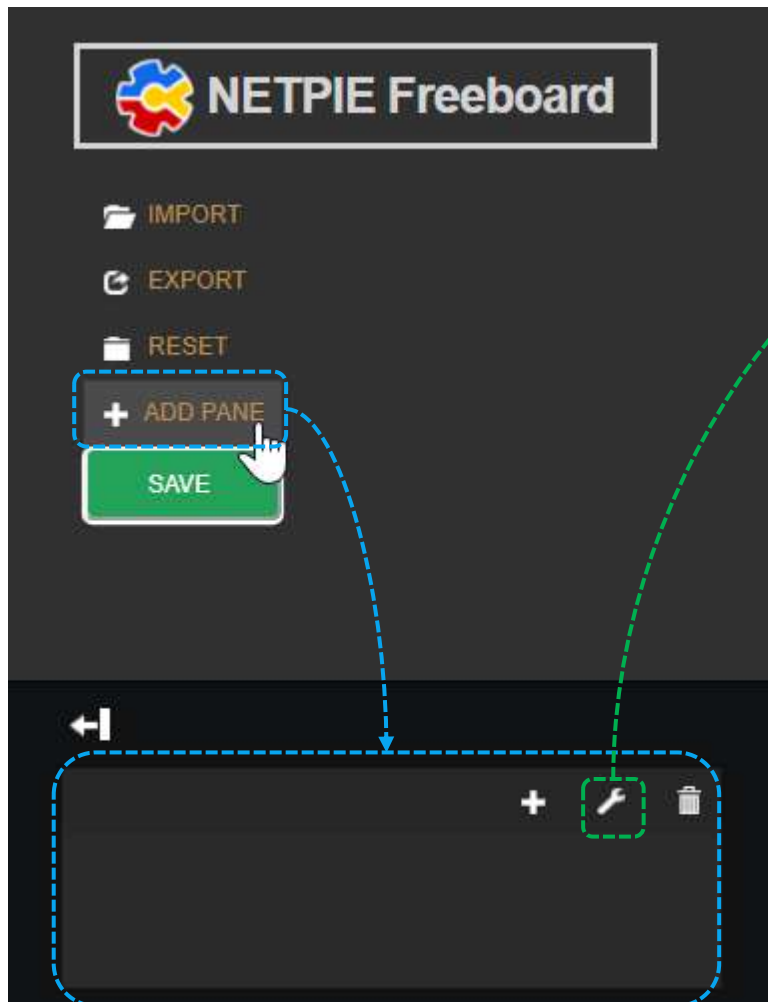
Enable :

คลิก เพื่อ Copy

Client ID

Token

ADD PANE in your Dashboard and Name your PANE



Setting your WIDGET (Example : Gauge)



WIDGET

TYPE: Gauge

TITLE: Temperature

VALUE:

UNITS: Celsius

MINIMUM: 0

MAXIMUM: 60

+ DATASOURCE JS EDITOR

SAVE CANCEL

ติดตั้งชื่อ

ระบุหน่วย

คลิก แล้วเลือก Gauge

กำหนดค่าต่ำสุด

กำหนดค่าสูงสุด

WIDGET

TYPE: Gauge

TITLE: Temperature

VALUE: datasources["Data_1"]["shadow"]["data101"]

UNITS: Celsius

MINIMUM: 0

MAXIMUM: 60

+ DATASOURCE JS EDITOR

SAVE CANCEL

datasources["Data_1"]

shadow

msg

datasources["Data_1"]

shadow

msg

datasources["Data_1"]["shadow"]

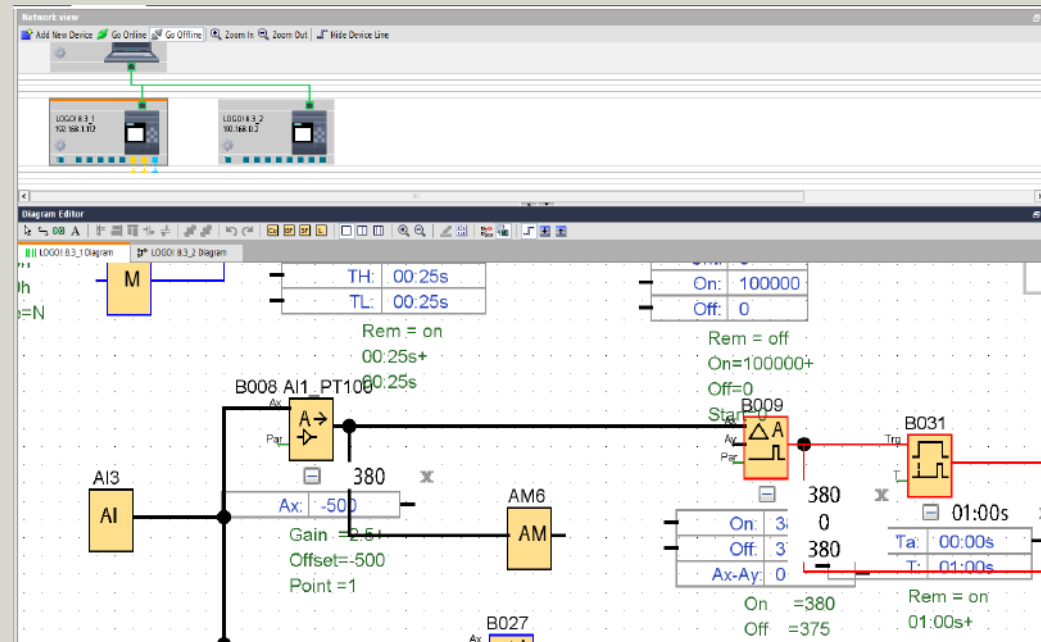
data101

datasources["Data_1"]["shadow"]["data101"]

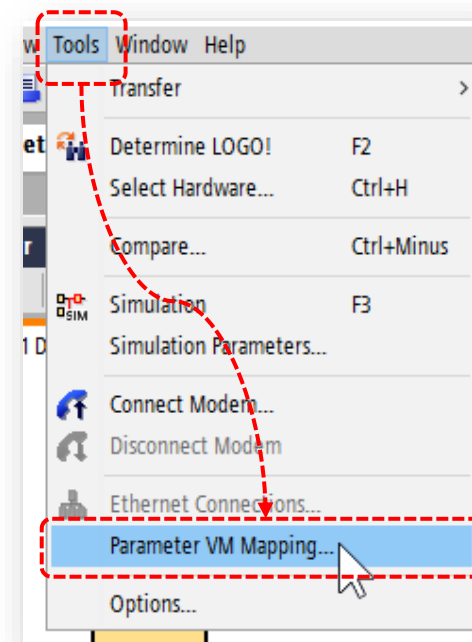
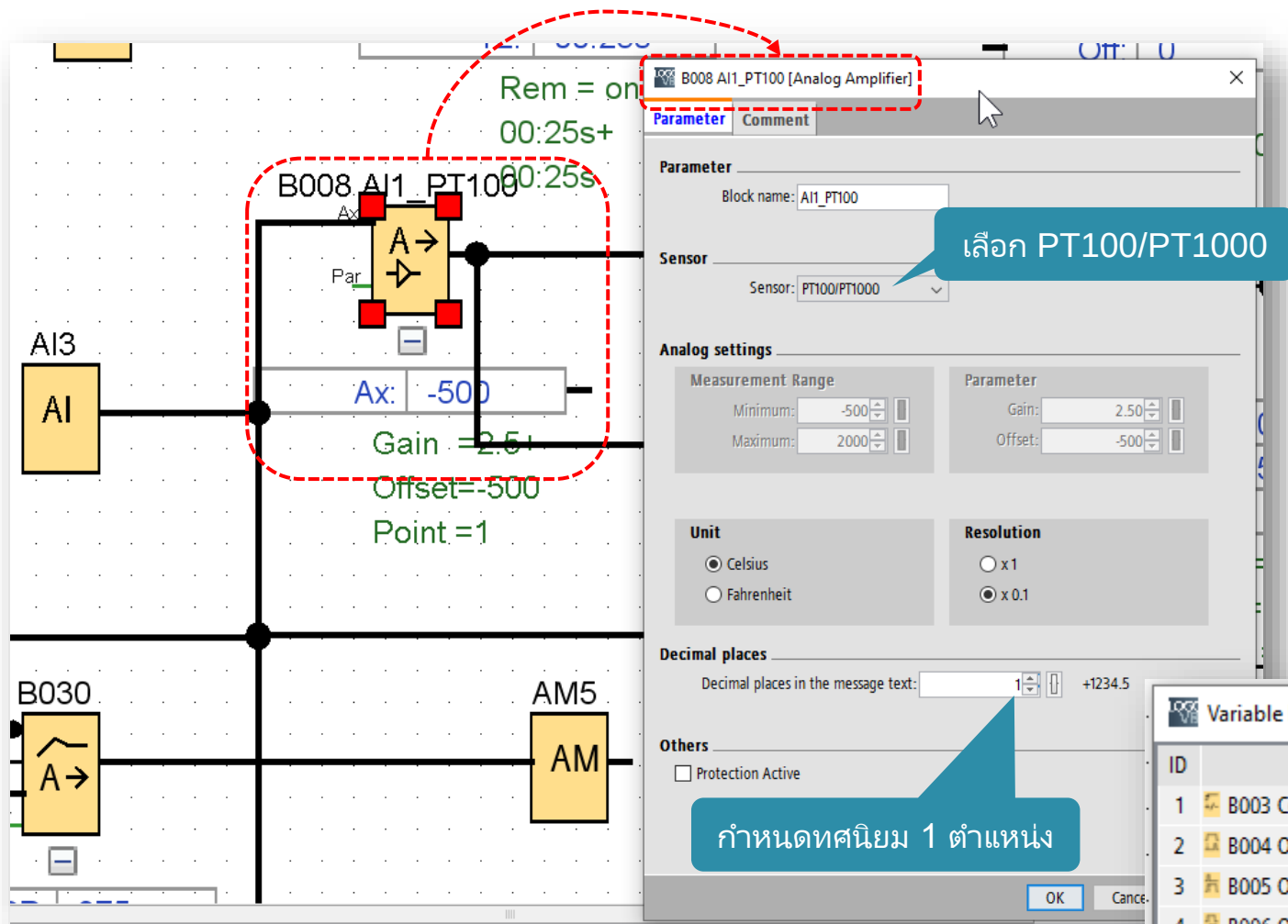
LOGO!8 + AI module (Temp) and S7 server connection



- 1) Read data from temperature input module (AI3)
- 2) Variable mapping (VW20)
- 3) Create s7 server connection



Read temp value : AI3 (PT100 sensor) Mapping Temp value to VW20 (internal memory of LOGO!8)

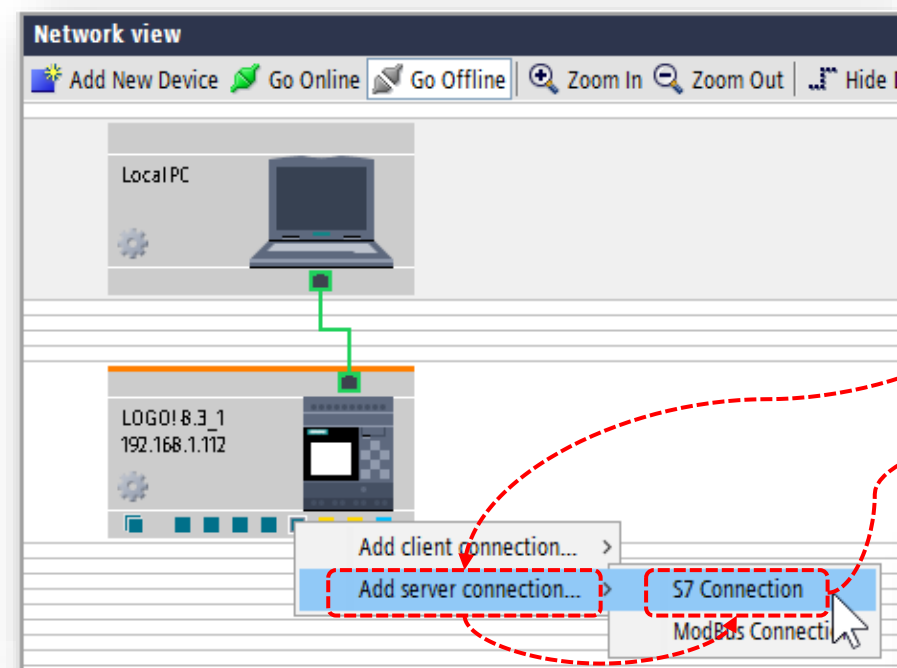


การอ่านค่าอุณหภูมิโดยใช้ Sensor ชนิด PT100 การตั้งค่าใน Analog Amplifier กำหนดการแสดงผลด้วยทศนิยม 1 ตำแหน่ง เช่น ค่าอุณหภูมิเท่ากับ 37.5 องศาเซลเซียส ค่าที่เก็บใน VW20 จะมีค่าเท่ากับ 375

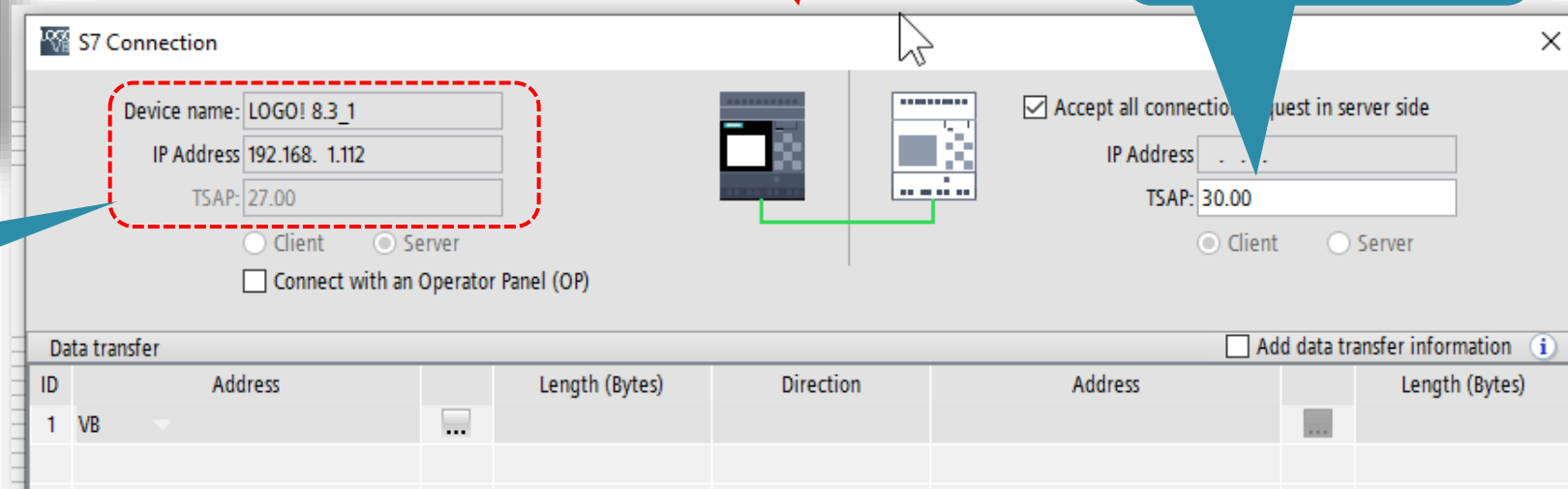
ID	Block	Parameter	Type	Address
1	B003 Count 3 [Up/Down counter]	Counter	DWord	10
2	B004 OnDelay4S [On-Delay]	Current Time	Word	14
3	B005 OffDelay3S [Off-Delay]	Current Time	Word	16
4	B006 On-Off Delay [On-/Off-Delay]	Current Time	Word	18
5	B008 AI1_PT100 [Analog Amplifier]	Ax, amplified	Word	20

VW20 เก็บข้อมูลชนิดจำนวนเต็ม

Create S7-Server connection for LOGO!8



หมายเลข TSAP
ของ LOGO!8 ตั้งไว้
ที่เช่น 27.00



SIMATIC IOT2050 : Flow based program with NODE-Red



1) Read Temperature data from SIMATIC LOGO!8

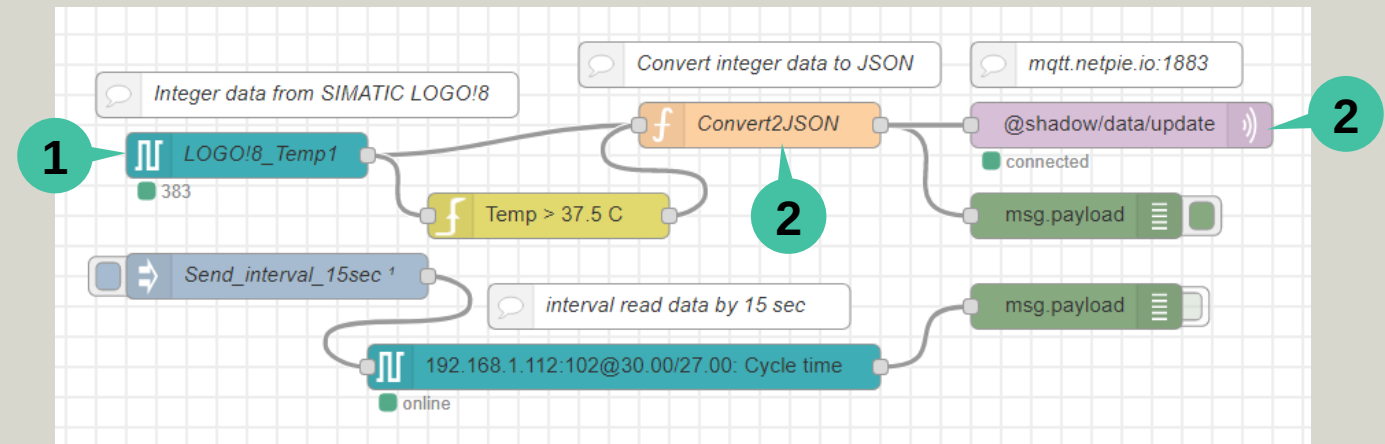
- Using s7 in node

2) Convert integer data to JSON

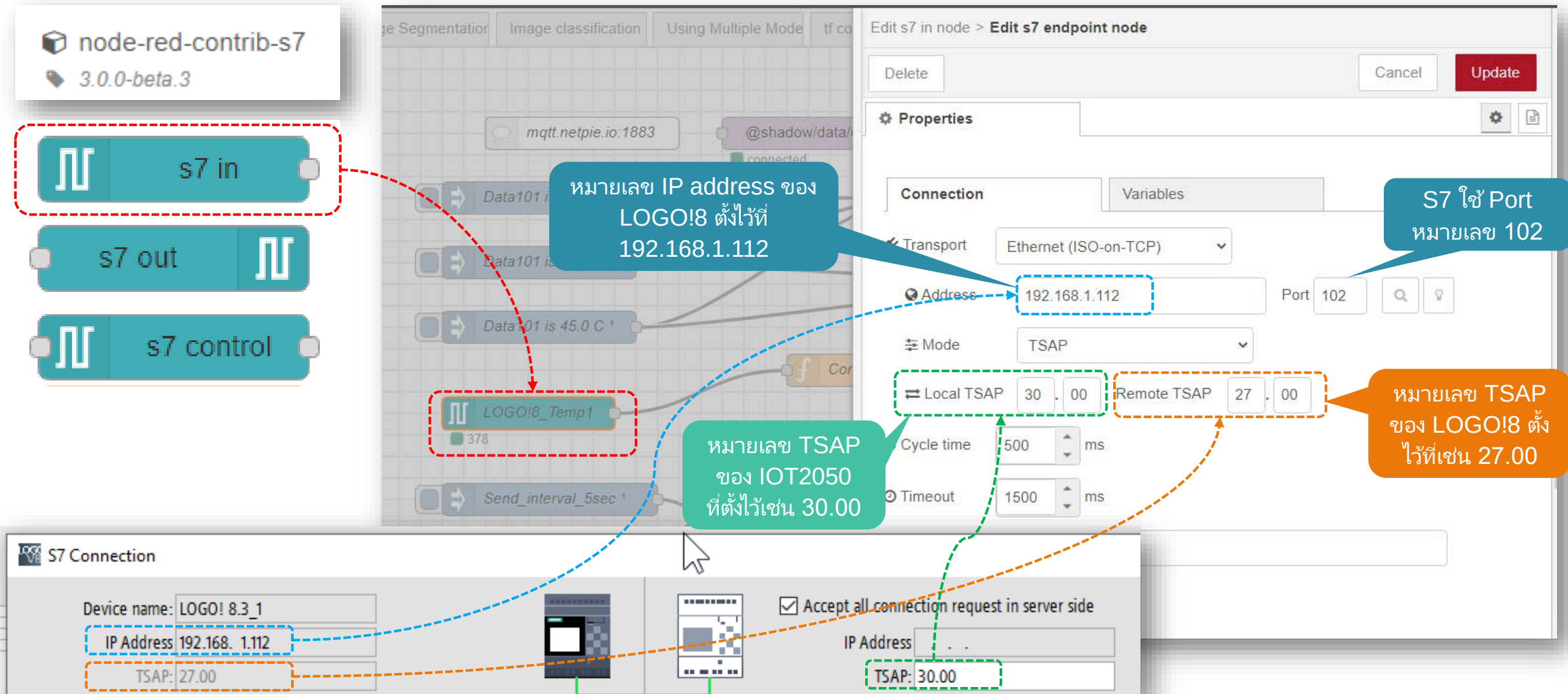
- Using function node

3) Ingest data to NETPIE

- Using mqtt out node



Reading Temp data from LOGO!8 to IOT2050 using NODE-RED



การตั้งค่า Variable สำหรับ S7 endpoint node VW20 ของ LOGO!8 ตรงกับ DB1,INT20 สำหรับ S7-Connection



Classification Using Multiple Mode If co Edit s7 in node > Edit s7 endpoint node

Delete Cancel Update

Properties

Connection Variables

Variable list

DB1,INT20 Temperature

DB1,INT30 Current

DB1,INT28 Voltage

การอ้างอิง VW20 ใน LOGO!8 ให้ระบุ DB1,INT20

Variable Memory Configuration

ID	Block	Parameter	Type	Address
1	B003 Count 3 [Up/Down counter]	Counter	DWord	10
2	B004 OnDelay45 [On-Delay]	Current Time	Word	14
3	B005 OffDelay35 [Off-Delay]	Current Time	Word	16
4	B006 On-Off Delay [On-/Off-Delay]	Current Time	Word	18
5	B008 AI1_PT100 [Analog Amplifier]	Ax, amplified	Word	20

VW20 เก็บข้อมูลชนิดจำนวนเต็ม

Edit s7 in node

Delete Cancel Done

Properties

PLC 192.168.1.112:102@30.00/27.00

Mode Single variable

Variable Temperature DB1,INT20

Temperature

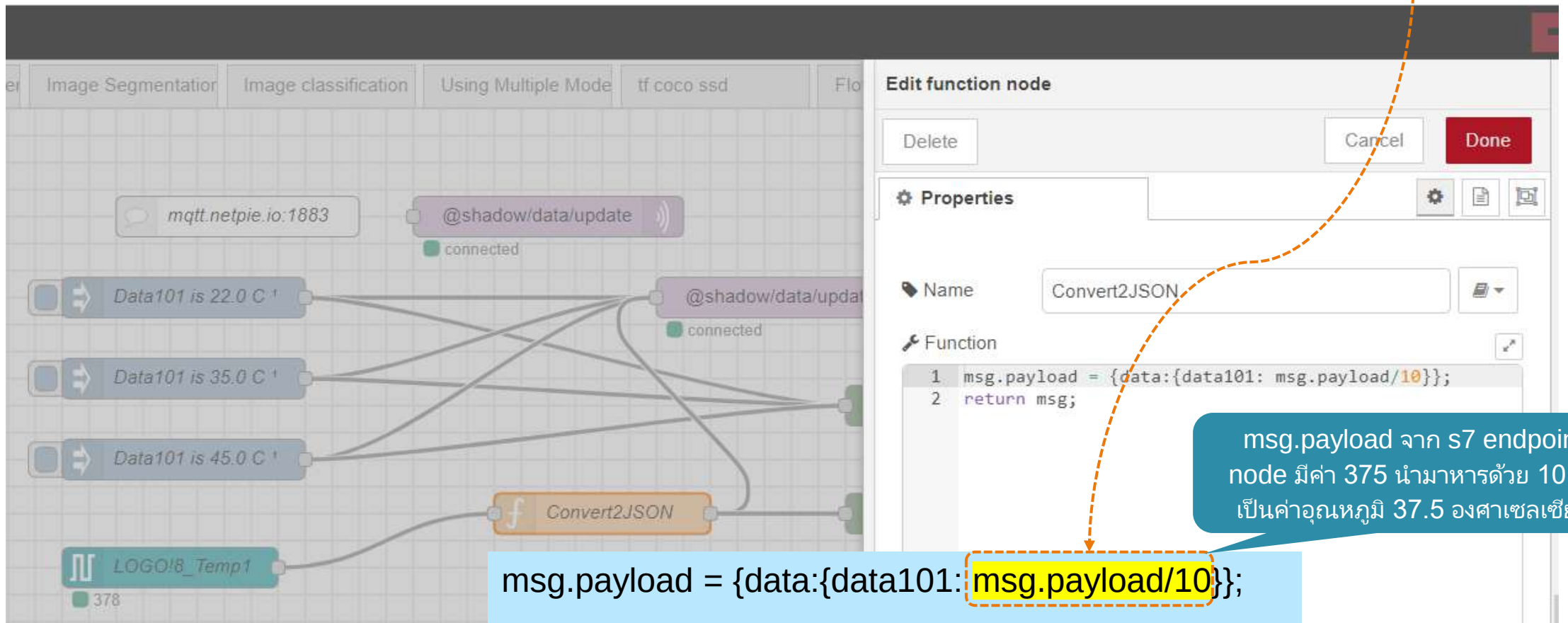
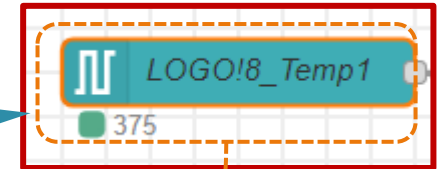
Current

Voltage

ใช้ Function แปลงข้อมูล INT จาก S7 endpoint node ให้เป็น JSON



ค่าอุณหภูมิจาก LOGO!8 เป็นข้อมูล
ชนิดจำนวนเต็ม 375 คือ อุณหภูมิ
37.5 องศาเซลเซียส



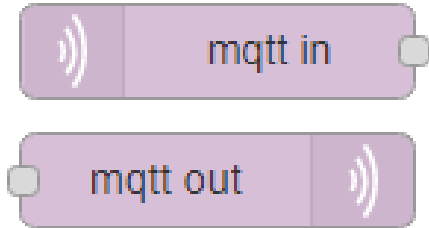
`msg.payload = {data:{data101: msg.payload/10}};`
`return msg;`

msg.payload จาก s7 endpoint
node มีค่า 375 นำมาหารด้วย 10 ได้
เป็นค่าอุณหภูมิ 37.5 องศาเซลเซียส

Send data to NETPIE using mqtt out node



network



The screenshot shows the Node-RED interface with the 'Edit mqtt out node' configuration panel open. The panel has a 'Delete' button, 'Cancel' and 'Done' buttons, and a 'Properties' section. The 'Properties' section includes fields for 'Server' (set to 'Add new mqtt-broker...'), 'Topic' (set to '@shadow/data/update'), 'QoS' (set to 0), 'Retain' (checked), and 'Name' (set to 'Name'). A blue callout box points to the 'Topic' field with the text '@shadow/data/update'. A yellow tip box at the bottom states: 'Tip: Leave topic, qos or retain blank if you want to set them via msg properties.'

Edit mqtt out node

Delete Cancel Done

Properties

Server Add new mqtt-broker...

Topic @shadow/data/update

QoS Retain

Name Name

Tip: Leave topic, qos or retain blank if you want to set them via msg properties.

การตั้งค่า mqtt-broker node สำหรับ NETPIE

Tab : Connection และ Security



Edit mqtt out node > Edit mqtt-broker node

1

ระบุ MQTT Server สำหรับ NETPIE ให้ใช้ mqtt.netpie.io หรือ broker.netpie.io และใช้ port หมายเลข 1883

Connection

Server mqtt.netpie.io Port 1883

☐ Enable secure (SSL/TLS) connection

Client ID

Keep alive time (s) 60 ☒ Use clean session

☐ Use legacy MQTT 3.1 support

Key

Client ID : Token : Secret :

Status : Online

Enable : ☒

คลิก เพื่อ Copy

3

Update

2

Security

Token

Secret

Deploy



Edit mqtt out node

4

Delete Cancel Done

Properties

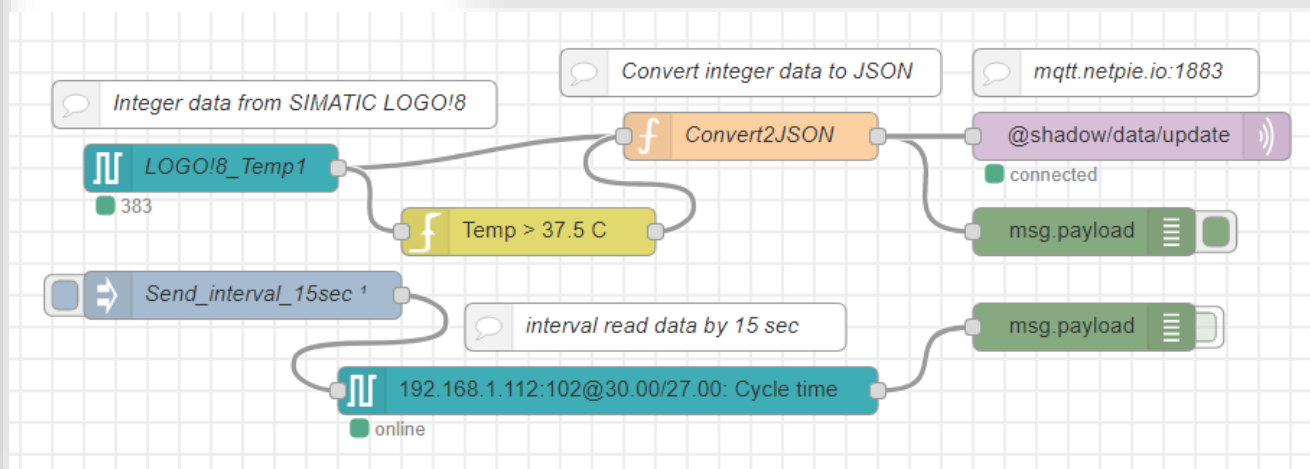
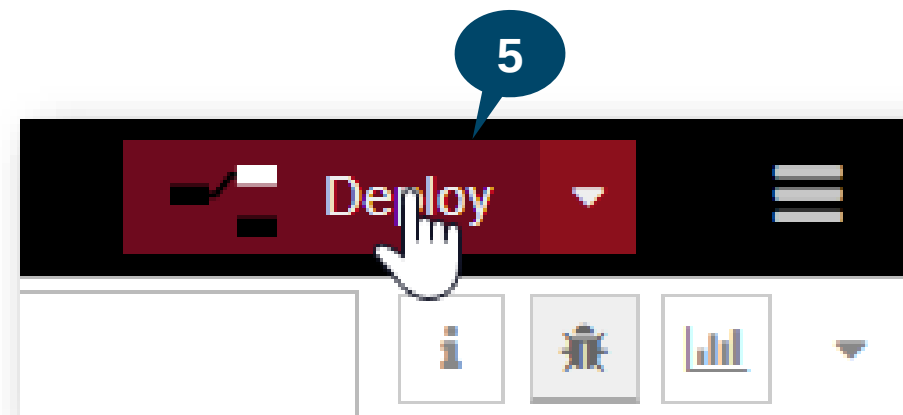
Server **abcd efg.....xyz**

Topic @shadow/data/update

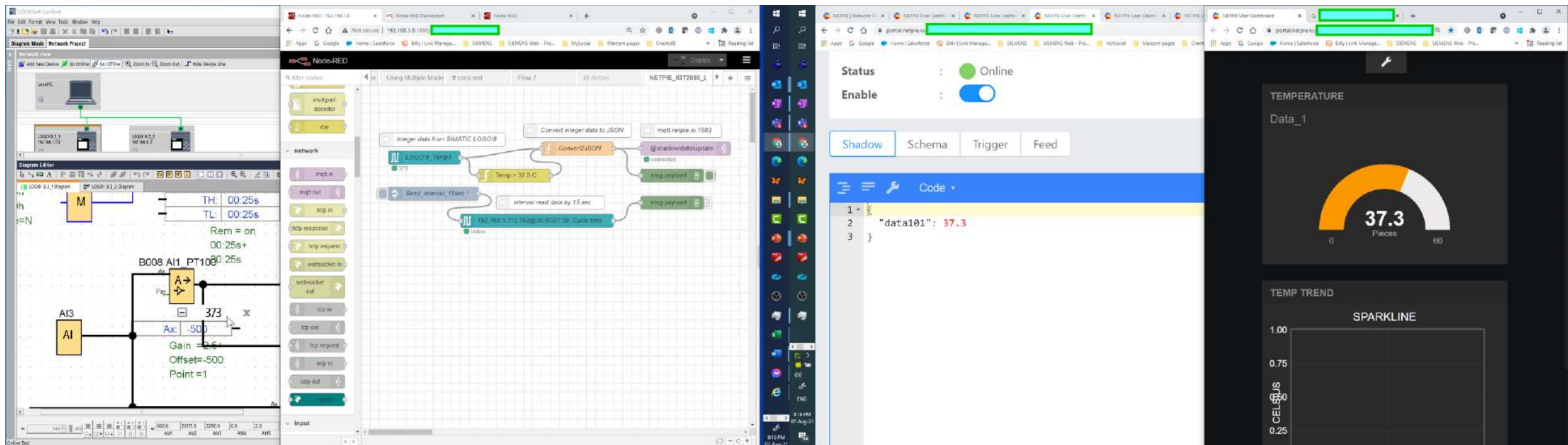
QoS 0 Retain true

Name Name

Tip: Leave topic, qos or retain blank if you want to set them via msg properties.



SIMATIC LOGO!8 <> SIMATIC IOT2050 <> NETPIE



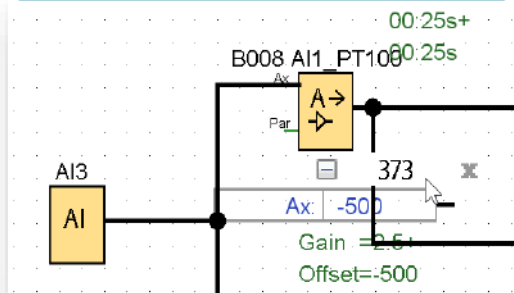
This block contains three screenshots illustrating the integration of SIMATIC LOGO!8, SIMATIC IOT2050, and NETPIE.

- Left Screenshot (SIMATIC Manager):** Shows the LAD editor for a SIMATIC LOGO!8 device. It features an AI3 input connected to a function block (A) with parameters: Gain = 2.5, Offset = -500, and Point = 1. The output is labeled 373.
- Middle Screenshot (Node-RED):** Shows a Node-RED flow titled "NETPIE_IOT2050_L". It includes a "LOGO!8_Temp1" node, a "Convert integer data to JSON" node, and a "mqtt.netpie.io:1883" node. A condition "Temp > 37.5 C" is also present.
- Right Screenshot (NETPIE Dashboard):** Shows the NETPIE User Dashboard. It displays the status as "Online" and "Enable". Below, there is a "Shadow" tab with a code editor showing a JSON payload:

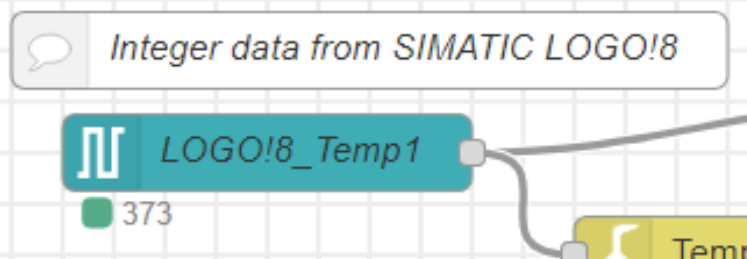
```
1 {  
2   "data101": 37.3  
3 }
```

. To the right, a gauge displays the temperature "37.3 Pieces" and a "TEMP TREND" sparkline chart.

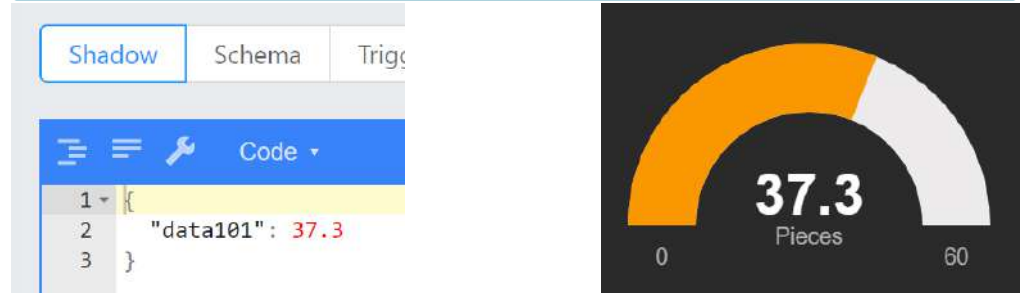
SIMATIC LOGO!8



SIMATIC IOT2050



NETPIE : Device shadow & Dashboard



A detailed view of the NETPIE dashboard. It shows the "Shadow" tab with a code editor displaying a JSON payload:

```
1 {  
2   "data101": 37.3  
3 }
```

. To the right, a gauge displays the temperature "37.3 Pieces" and a "TEMP TREND" sparkline chart.

Facebook group “Machine Automation”



Machine Automation

Private group



Interacting as yourself

About

Discussion

Announcements

Members

Events

Videos

Photos

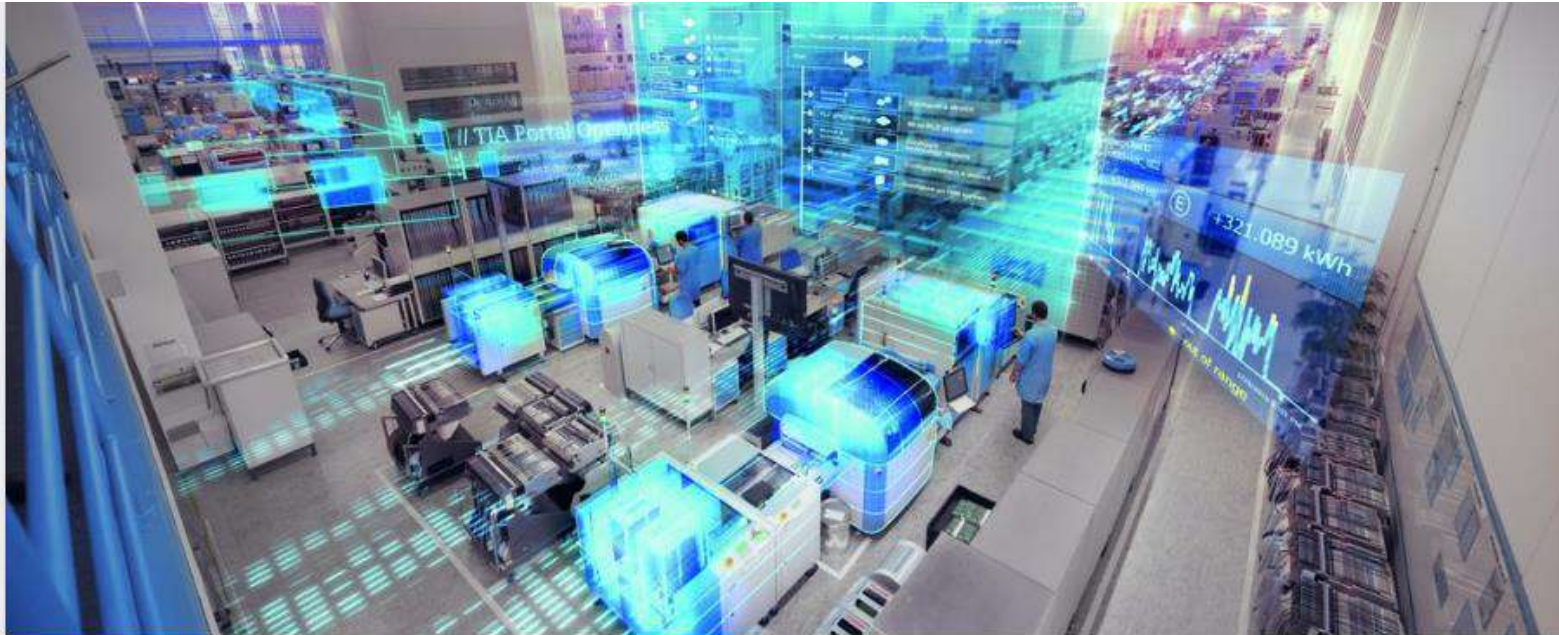
Files

Group Insights

Watch Party

Moderate Group

Group Quality



Joined ▾

✓ Notifications

➦ Share

⋮ More



Write Post



Photo/Video



Live Video



More



Write something....

GROUP BY

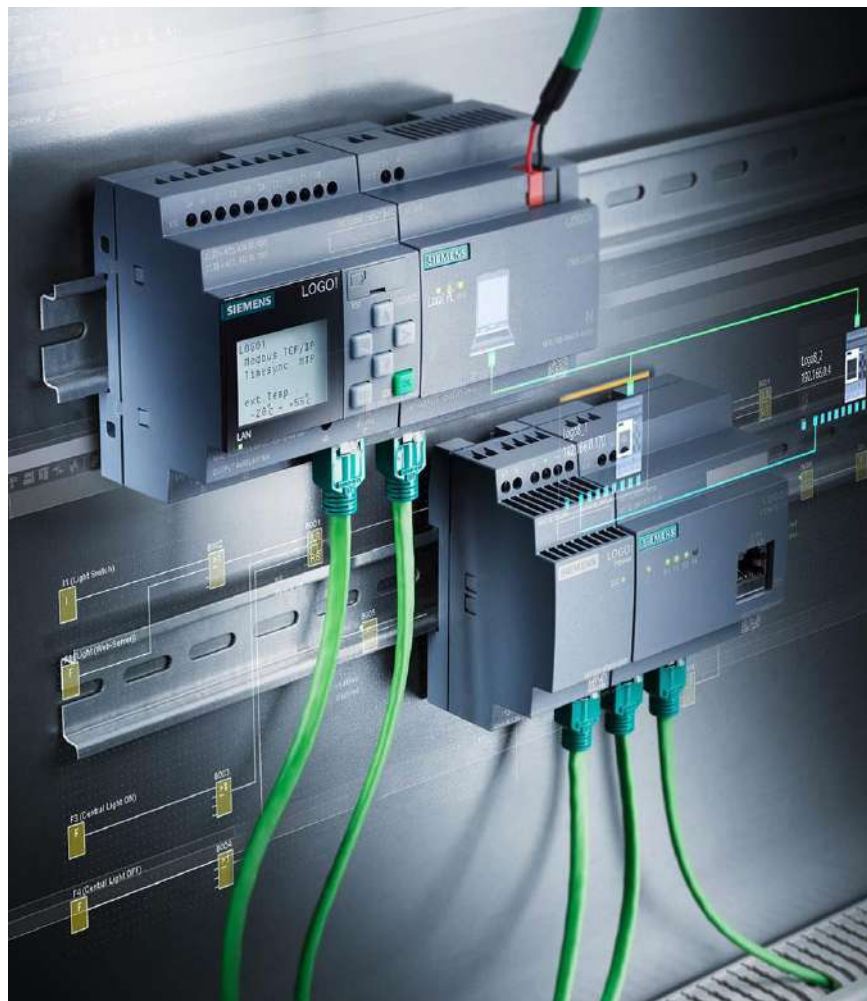


Siemens Thailand

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